Research activity

Height of a point on a wheel — 35 min.

A wheel turns at constant speed in a vertical plane. Its radius is 2 m and its centre is 3 m above the ground. A point on the wheel is at its lowest position at time $t = 0$. The wheel makes one revolution in 8 s. Its height is modelled by $h(t) = 3 - 2\cos\left(\frac{\pi t}{4}\right)$, where t is in seconds and $h(t)$ in metres.

1. Check that the model gives the stated initial height, a period of 8 s and the expected minimum and maximum heights.
2. Calculate $h'(t)$, then study the variations of h on $[0;8]$.
3. Determine all times $t \in [0;8]$ for which the height is 4 m.
4. Determine the total time, during this revolution, for which the point is at least 4 m above the ground.